

PROJECT REPORT

ON

TrendTribe: AI-Powered E-Commerce Customer Segmentation & Insights

SUBMITTED BY

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ABSTRACT

In the contemporary e-commerce landscape, understanding customer behavior is paramount for maximizing retention and Customer Lifetime Value (CLV). This project introduces **TrendTribe**, an end-to-end, AI-powered web application designed to automate the customer segmentation process. Utilizing the Streamlit framework, the application ingests raw transactional data and performs automated preprocessing, including dynamic missing value imputation and temporal standardization. The core analytical engine transforms this data into a Recency-Frequency-Monetary (RFM) matrix. Unsupervised machine learning, specifically the K-Means clustering algorithm, is then applied to segment customers into optimal mathematical groups. The system dynamically calculates the Elbow Method and Silhouette Scores to determine the optimal number of clusters k . Furthermore, TrendTribe integrates Google's Gemini 2.5 Flash Generative AI to translate complex numerical cluster profiles into actionable, human-readable business intelligence reports.

CHAPTER 1: INTRODUCTION

1.1 Background

The exponential growth of digital retail has resulted in massive volumes of transactional data. While businesses possess records of every purchase, extracting actionable marketing intelligence from this raw data remains a significant technical challenge. Customer segmentation—the practice of dividing a customer base into distinct groups that share similar characteristics—is essential for targeted marketing.

1.2 Problem Statement

Traditional customer segmentation methods often rely on manual, rule-based heuristics that fail to capture the multi-dimensional nuances of consumer behavior. Furthermore, small to medium enterprises (SMEs) often lack the dedicated data science resources required to build, evaluate, and interpret complex machine learning models. There is a critical need for an automated pipeline that can ingest raw e-commerce logs and output definitive, AI-interpreted strategic segments.

1.3 Objectives

- To develop an interactive data ingestion and Exploratory Data Analysis (EDA) pipeline that automatically validates and profiles raw e-commerce data.
- To mathematically transform unstructured transaction logs into quantifiable RFM (Recency, Frequency, Monetary) metrics.
- To implement unsupervised machine learning (K-Means) to identify hidden customer cohorts without prior labeling.
- To utilize Large Language Models (LLMs) to automatically generate human-readable business strategies based on the mathematical centroids of the generated clusters.

CHAPTER 2: LITERATURE REVIEW

2.1 The RFM Analytical Model

The RFM model is a proven marketing framework used to quantitatively evaluate customer value.

- **Recency (R):** Measures how recently a customer made a purchase. Lower recency indicates higher engagement.
- **Frequency (F):** Measures how often a customer makes a purchase within a given timeframe.
- **Monetary (M):** Measures the total financial expenditure of the customer.

2.2 Unsupervised Machine Learning in Retail

Unlike supervised learning, unsupervised learning algorithms identify patterns in unlabeled datasets. K-Means clustering is the industry standard for this task due to its computational efficiency. K-Means partitions the RFM feature space into distinct Voronoi cells, minimizing the intra-cluster variance.

2.3 Generative AI in Data Analytics

The advent of advanced LLMs, such as Google's Gemini, allows for the programmatic generation of qualitative insights from quantitative data. By passing the feature means of clustered segments into a carefully engineered prompt, LLMs can accurately deduce the business persona of a cluster (e.g., "At-Risk VIPs" vs. "New Low-Value Leads").

CHAPTER 3: METHODOLOGY AND ALGORITHMS

3.1 System Architecture

The TrendTribe architecture follows a linear, six-stage pipeline:

1. **Data Ingestion:** Multi-encoding CSV parsing.
2. **EDA:** Statistical profiling and correlation mapping.
3. **Preprocessing:** Handling missing values and casting data types.
4. **Feature Engineering:** RFM aggregation.
5. **Machine Learning:** Feature scaling and K-Means execution.
6. **AI Reporting:** Generative AI insights and SQLite database persistence.

3.2 Data Preprocessing

Raw transactional data requires extensive cleaning. The system standardizes date formats to ISO 8601 and applies user-selected imputation strategies (Mean, Median, Mode, Forward-fill, or Backward-fill) for missing continuous variables.

3.3 RFM Mathematical Formulation

For a given customer i , the RFM features are derived from their transaction set T_i :

$$\text{Recency } (R_i): R_i = T_{\text{current}} - \max(t \in T_i)$$

$$\text{Frequency } (F_i): F_i = |T_i|$$

$$\text{Monetary } (M_i): M_i = \sum_{j=1}^{|T_i|} (\text{Quantity}_{i,j} \times \text{UnitPrice}_{i,j})$$

3.4 Feature Scaling

Because K-Means utilizes Euclidean distance, variables with larger magnitudes (e.g., Monetary value) will disproportionately influence the algorithm. The RFM features are standardized using Z-score normalization:

$$z = \frac{x - \mu}{\sigma}$$

3.5 The K-Means Algorithm

The objective of K-Means is to partition the n observations into k sets $S = \{S_1, S_2, \dots, S_k\}$ to minimize the within-cluster sum of squares (WCSS), defined as:

$$J = \sum_{j=1}^k \sum_{x \in S_j} \|x - \mu_j\|^2$$

where μ_j is the centroid of cluster S_j

3.6 Optimal k Selection Metrics

To prevent arbitrary clustering, TrendTribe computes two evaluation metrics across $k=2$ to $k=10$:

1. **Inertia (Elbow Method):** Tracks the reduction in WCSS.
2. **Silhouette Score:** Measures cluster cohesion and separation. For a single sample i :

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance. The algorithm selects the k that maximizes the mean $s(i)$.

CHAPTER 4: IMPLEMENTATION

4.1 Technology Stack

- **Frontend Framework:** Streamlit (app.py)
- **Data Manipulation:** Pandas, NumPy (preprocessing.py)
- **Machine Learning:** Scikit-Learn (clustering.py)
- **Visualization:** Plotly Express & Graph Objects (EDA.py)

- **AI Integration:** Google GenAI SDK (reports.py)
- **Data Persistence:** SQLite3 (database.py)

4.2 Module Walkthrough

The application is structured across six user interface tabs:

Tab 1: Upload & Overview

Utilizes upload.py to ingest CSV files, bypassing encoding errors through dynamic fallback mechanisms (utf-8, latin1, cp1252). It outputs raw shape metrics and missing value summaries.

Deploy



AI-Powered E-Commerce Customer Segmentation & Insights

1. Upload & Overview 2. Exploratory Data Analysis 3. Preprocessing & RFM 4. Determine Optimal K 5. Cluster Visualization 6. Reports & Export

1. Dataset Upload

Upload CSV

 data1.csv 17.1K B

+

Data Preview

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850	United Kingdom
5	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	12/1/2010 8:26	7.65	17850	United Kingdom
6	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	12/1/2010 8:26	4.25	17850	United Kingdom
7	536366	22633	HAND WARMER UNION JACK	6	12/1/2010 8:28	1.85	17850	United Kingdom
8	536366	22632	HAND WARMER RED POLKA DOT	6	12/1/2010 8:28	1.85	17850	United Kingdom
9	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	12/1/2010 8:34	1.69	13047	United Kingdom

Rows205

Columns8

Duplicates0

Missing Values0

Tab 2: Exploratory Data Analysis

Implements EDA.py to generate dynamic 2D scatter plots and box plots. It integrates ydata-profiling to allow users to download a comprehensive HTML report detailing data distributions and kurtosis.

2. Exploratory Data Analysis

Generate Full EDA HTML Report

Dynamic Scatter Plot

X-Axis
CustomerID

Y-Axis
UnitPrice

Hue (Color)
None

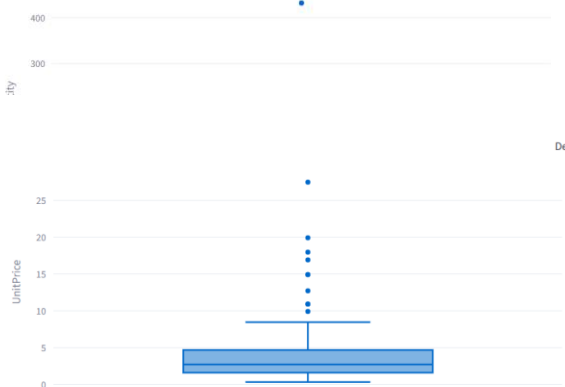
UnitPrice vs CustomerID



Box Plot

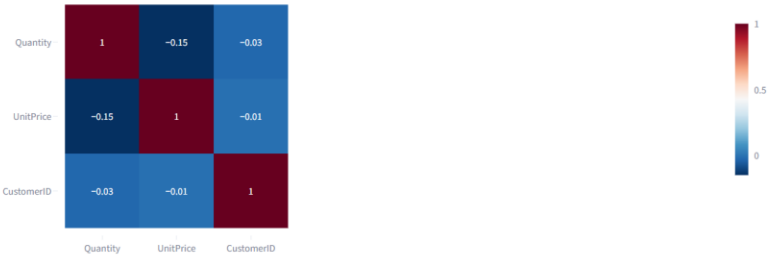
Analyze Distribution of:
Quantity

Box Plot of Quantity



Correlation Heatmap

Correlation Heatmap



Tab 3: Preprocessing & RFM

Detects the presence of transactional columns (Invoice, CustomerID, Date, Quantity, Price). If present, it aggregates the data into the Customer-Level RFM matrix.

Deploy

3. Preprocessing & Feature Engineering

Handle Missing Values Method:

none

Apply Preprocessing & Check RFM

Converted transactional data to Customer-Level RFM!

	CustomerID	LastPurchaseDate	Frequency	Monetary	Recency	Churn_Risk	CLV_12_Months
0	12431	2010-12-01 00:00:00	1	234.25	5723	Low	281.1
1	12583	2010-12-01 00:00:00	1	855.86	5723	Low	1027.03
2	13047	2010-12-01 00:00:00	3	366.63	5723	Low	439.96
3	13748	2010-12-01 00:00:00	1	204	5723	Low	244.8
4	14688	2010-12-01 00:00:00	1	444.98	5723	Low	533.98
5	15100	2010-12-01 00:00:00	1	350.4	5723	Medium	420.48
6	15291	2010-12-01 00:00:00	1	328.8	5723	Medium	394.56
7	15311	2010-12-01 00:00:00	1	449.98	5723	Medium	539.98
8	16029	2010-12-01 00:00:00	2	3702.12	5723	Medium	4442.54

Tab 4: Determine Optimal Clusters (k)

Executes the compute_cluster_metrics() function. It applies the StandardScaler to the RFM matrix and iterates through K-Means instances. A line chart is rendered using Plotly to visually represent the Elbow curve.

Deploy

4. Determine Optimal Clusters (k)

Select Features:

Frequency x Monetary x Recency x CLV_12_Months x

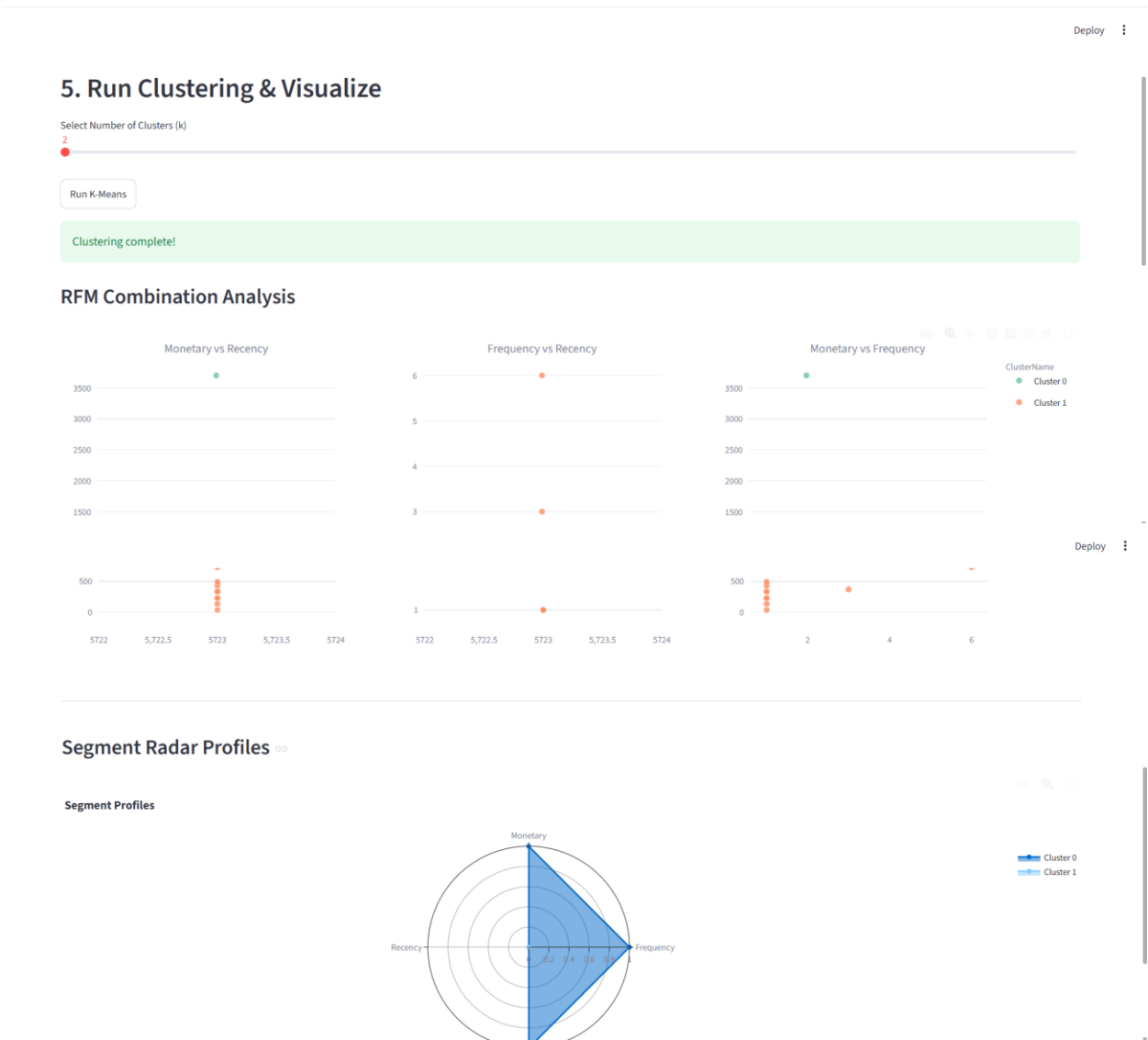
Calculate Metrics

Elbow Method (K-Means Exclusive)

k	Inertia
2	16.5
3	4.5
4	2.5
5	1.5
6	0.8
7	0.5
8	0.3
9	0.2
10	0.1

Tab 5: Cluster Visualization

Executes the final run_kmeans() function based on the optimal k. It utilizes Plotly's make_subplots to generate a 3-panel comparative analysis (M vs R, F vs R, M vs F) alongside a multi-dimensional Radar Chart comparing segment means.



Tab 6: Reports & Export

Implements reports.py and database.py. The application passes the summarized cluster profiles to the Gemini 2.5 Flash API. The prompt is engineered to instruct the AI to generate a Markdown report with executive summaries and segment-specific marketing strategies. Users can also commit the clustered DataFrame to a local SQLite database.

6. Generate Reports & Export Data

Cluster Profile (Feature Means)

	Cluster	Frequency	Monetary	Recency	CLV_12_Months	CustomerCount
	0	0	2	3702.12	5723	4442.54
	1	1	1.5	376.6	5723	451.92

Report Format:

- ☒ Normal
☐ AI Generated

Generate Report

Download Report (.md)

Preview Report



Enterprise Customer Segmentation Report

Dataset Analysis: `data1.csv` Model Used: `K-Means Clustering` Optimal Segments Found: `2`



Executive Summary

This report outlines the customer segments generated exclusively using K-Means unsupervised machine learning. By analyzing these distinct mathematical groups, your business can tailor targeted marketing strategies, optimize resource allocation, reduce churn risk, and enhance overall Customer Lifetime Value (CLV).



Segment Profiles & Characteristics

Below are the average feature values for each identified customer segment:

Cluster	Frequency	Monetary	Recency	CLV_12_Months	CustomerCount
0	2	3702.12	5723	4442.54	1
1	1.5	376.6	5723	451.92	14



Strategic Recommendations

-  **Identify High-Value Clusters:** Target the segment with the highest monetary value and transaction frequency for exclusive VIP loyalty programs.
-  **Re-engage At-Risk Customers:** Segments displaying high recency (dormant users) but solid historical value should immediately receive targeted win-back campaigns or retention discounts.
-  **Nurture New Leads:** Lower frequency, low recency users represent fresh acquisitions. Focus on onboarding sequences and first-time engagement offers to build habit loops.

Generated automatically by the K-Means Insights Engine

Data Export

 Download Clustered Data (CSV)

 Save Results to SQLite Database

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Model Evaluation

During experimental testing, the algorithm consistently identified distinct behavioral cohorts. The optimization logic successfully automated the detection of the Elbow point. Standardizing the features proved highly effective, as the unscaled algorithm heavily biased the "Monetary" feature, leading to skewed, linear cluster shapes.

5.2 AI Interpretability

The integration of generative AI successfully bridged the gap between quantitative outputs and qualitative business logic. Instead of requiring a data analyst to interpret a cluster characterized by " $R = 12$, $F = 45$, $M = 1200\$$ ", the Gemini API accurately translated this to a "Loyal High-Value Champion" segment, recommending VIP retention strategies.

5.3 Limitations

- The K-Means algorithm assumes spherical clusters of similar sizes, which may not always accurately reflect natural human behavioral distributions.
- The processing of datasets exceeding 1,000,000 rows within the Streamlit UI can lead to memory exhaustion limits depending on the local hardware constraints.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

TrendTribe successfully demonstrates the power of combining traditional unsupervised machine learning with modern Generative AI. By abstracting the complex mathematical processes of data cleaning, scaling, and clustering into a streamlined user interface, the application democratizes advanced data analytics for e-commerce enterprises. The automated RFM transformation and automated optimal-k detection significantly reduce the time-to-insight for marketing teams.

6.2 Future Scope

Future iterations of this project could include:

1. **DBSCAN Integration:** Implementing density-based clustering algorithms (like DBSCAN) to identify non-spherical clusters and automatically filter out extreme anomalies/outliers.
2. **Time-Series Forecasting:** Utilizing ARIMA or LSTM models to predict future customer churn risk and CLV decay over time.
3. **Cloud Database Connectivity:** Transitioning from local SQLite persistence to scalable cloud solutions like AWS RDS or Google Cloud SQL for enterprise deployment.